IRREDUCIBLE COMPONENTS BECOME GEOMETRICALLY IRREDUCIBLE AFTER A FINITE SEPARABLE EXTENSION OF THE BASE FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the author's own 'and we win'; no step is referred to the reader.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-finite-extension-geometrically-irreducible-components is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Finiteness of the orbits of the Galois action on irreducible components. Same chapter, Lemma lemma-galois-action-irreducible-components-locally-finite-type part (3), invoked at proof lines 1859-1863. Supplies that the components over the separable closure form a finite list.
- Definability of every irreducible component of the base change over a finite extension. Same chapter, Lemma lemma-galois-action-irreducible-components-locally-finite-type part (2), invoked at proof lines 1863-1867. Supplies the finite extension over which each component descends.

Source reference. Original lemma 04KZ: sources/stacks-varieties/pinned-original-source.tex, source lines 1789–1789. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let $X \to \mathrm{Spec}(k)$ be locally of finite type. Assume X has finitely many irreducible components. Then there exists a finite separable extension k'/k such that every irreducible component of $X_{k'}$ is geometrically irreducible over k'.*

Proof. Let $\overline{k}$ be a separable algebraic closure of k. The assumption that X has finitely many irreducible components combined with Lemma 04KZ (3) shows that $X_{\overline{k}}$ has finitely many irreducible components $\overline{T}_1, \ldots, \overline{T}_n$. By Lemma 04KZ (2) there exists a finite extension $\overline{k}/k'/k$ and irreducible components $T_i \subset X_{k'}$ such that $\overline{T}_i = T_{i,\overline{k}}$ and we win. □

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